Smart Manufacturing IoT-Based Predictive Maintenance & Anomaly Detection

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Outline

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Problem & Motivation

- **Context:** IoT sensors continuously monitor factory equipment (temperature, vibration, pressure, humidity)
- **Problem:** Most facilities still use reactive or scheduled maintenance
  - Unplanned downtime costs hundreds of thousands of dollars per hour
  - Unnecessary servicing wastes resources
- **Goal:** Predict maintenance needs from sensor data before failures occur
- **Secondary goal:** Detect anomalies without labeled data (unsupervised)

Dataset Overview

Smart Manufacturing IoT-Cloud Monitoring Dataset (Kaggle)

- 100,000 records, 50 machines
- 1-minute intervals over 69 days
- 13 columns (5 sensor, 3 derived, 2 operational, 2 ID, 1 target)
- No missing values

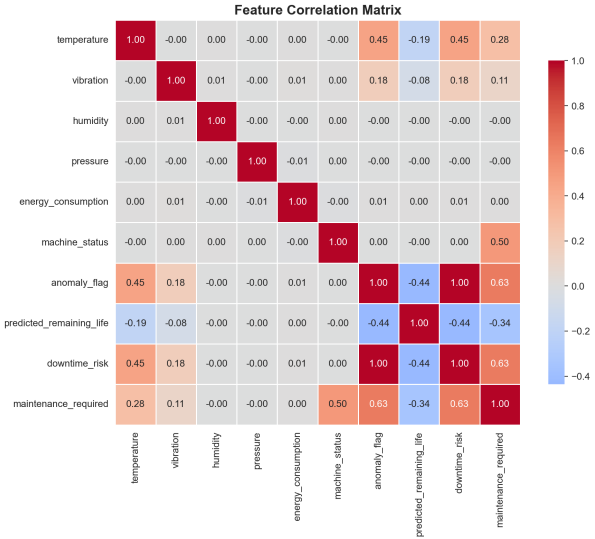
Class distribution:

- No maintenance: 80.3%
- Maintenance required: 19.7%

Key correlations with target:

- Temperature: $r = 0.283$
- Vibration: $r = 0.113$

Correlation Heatmap



Weak individual correlations suggest **non-linear feature combinations** are important.

Preprocessing

1. Dropped non-predictive identifiers (`timestamp`, `machine_id`)
2. Label-encoded `failure_type`
3. Retained 10 features (5 sensor + 2 operational + 3 derived)
4. Stratified 80/20 train-test split (80k / 20k samples)
5. StandardScaler fitted **only on training data** (prevents leakage)
6. `class_weight='balanced'` for imbalanced classes

ML Techniques Applied

Supervised:

1. **Logistic Regression** — interpretable baseline with cross-validation
2. **Random Forest** — non-linear ensemble with GridSearchCV tuning
3. **ANN (TensorFlow/Keras)** — deep non-linear modeling with Early Stopping

Unsupervised / Reduction:

4. **PCA** — dimensionality reduction, variance analysis
5. **Isolation Forest** — anomaly detection without labels

Evaluation metrics:
Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrices

ANN Architecture

Input (10) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dense(16, ReLU) $\rightarrow$ Dense(1, sigmoid)

- 3,329 trainable parameters
- Adam optimizer (lr $= 0.001$), binary crossentropy
- Early Stopping (patience $= 10$) with best-weight restoration
- Balanced class weights, batch size 256, 20% validation split

Model Comparison

Model	Acc.	Prec.	Rec.	F1	AUC
Naive Baseline	$\sim$0.80	—	—	—	—
Logistic Regression	0.9840	1.0000	0.9190	0.9578	0.9903
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
Neural Network	0.9963	0.9828	0.9987	0.9907	1.0000
LR + PCA (8 PCs)	0.9775	1.0000	0.8855	0.9393	0.9826
Isolation Forest	0.9381	0.8335	0.8566	0.8449	0.9604

- All supervised models significantly outperform the naive baseline ($\sim$0.80)
- RF achieves perfect scores — but warrants scrutiny (possible feature leakage)
- ANN provides the best precision–recall trade-off for maintenance

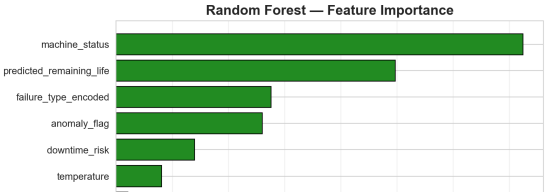
Key Results: Logistic Regression & Random Forest

Logistic Regression:

- Perfect precision (1.0000) — **zero** false alarms
- Recall = 0.9190 (319 missed out of 3,939)
- CV F1: 0.9578 ± 0.0051 (stable)
- Conservative boundary: only flags when highly confident, so missed failures could cause unplanned downtime

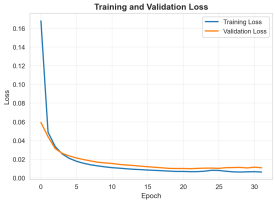
Random Forest (tuned):

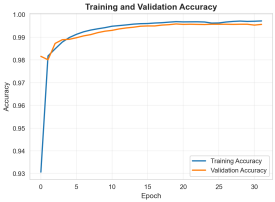
- All metrics = 1.0000 (perfect)
- Best params: 100 trees, depth 10
- Top features are **derived/operational**, not raw sensors — suggests data leakage inflates performance



Key Results: Neural Network

- F1 = 0.9907, Recall = 0.9987
- Only **5 missed failures** (vs. 319 for LR) — a plant running this model catches nearly every impending failure
- 69 false alarms trigger only preventive inspections, far cheaper than one unplanned breakdown
- Early Stopping at epoch ∼22 confirmed the model learns essential patterns early





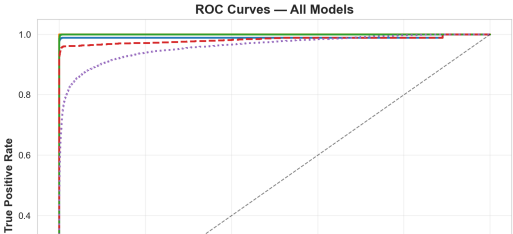
PCA & Isolation Forest

PCA:

- 8/10 components for 95% variance
- PC1: 26.0%, PC2: 15.3%
- LR+PCA F1 = 0.9393 (ΔF1 = -0.0185)
- Sensors encode largely distinct information; compression yields minimal benefit here

Isolation Forest (unsupervised):

- F1 = 0.8449, AUC = 0.9604
- 674 false positives waste technician time; 565 missed failures risk downtime
- Still viable as a first line of defense when labeled failure data is unavailable



Strengths & Limitations

Strengths:

- Five techniques across model families
- Rigorous evaluation (stratified split, CV, multiple metrics, naive baseline)
- Data leakage prevention
- Hyperparameter tuning & regularization
- Actionable feature importance

Limitations:

- Synthetic dataset — may not reflect real-world noise
- Derived features may encode the target (leakage concern)
- No temporal modeling (records treated independently)
- Single dataset — generalizability unvalidated

Conclusion & Future Work

Key findings:

- ANN achieved the best precision–recall trade-off (F1 = 0.9907, only 5 missed)
- RF's perfect scores likely reflect feature leakage, not true generalization
- Isolation Forest provides viable label-free detection (AUC = 0.9604)
- PCA confirms only moderate feature redundancy

Future work:

- LSTM / GRU for temporal degradation patterns
- Retrain without derived features to assess raw sensor performance
- SHAP values for instance-level explainability
- Validate on real-world industrial datasets

Thank You

Questions?

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